Exam Date & Time: 04-Jan-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH MAKEUP EXAMINATIONS, JANUARY 2024

OPERATING SYSTEMS [DSE 3153]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Answer the following based on the given scenarios:

A) a) Consider the **Table Q1** which shows five processes and two queues under multilevel queue scheduling. Q_no denotes the queue of the process, priority of queue 1 is greater than queue 2. Queue 1 uses round robin with time quantum of two units and queue 2 uses first come first serve scheduling algorithm.

b) Consider all the five processes are in same queue (say, Q_no = 2) and it follows preemptive shortest job first scheduling.

c) Consider all the five processes are in same queue (say, Q_no = 2) and it follows preemptive priority scheduling.

Table Q1:

PID	P1	P2	P3	P4	P5
AT	0	3	5	0	2
BT	6	4	1	5	2
Q_no	1	1	2	2	1
Priority	3	4	5	2	1

(5)

PID: Process ID, AT: Arrival time, BT: Burst time, Q_no: Queue no in which process reside, Priority: Lower number have the highest priority

For each of the above questions (a), (b), and (c) illustrate the working of the mentioned CPU Scheduling Algorithms with help of a Gantt chart and calculate average waiting time.

B) With a suitable diagram, explain Process Control Block and its components. (3)

C) Why is it important for the scheduler to distinguish I/O-bound programs from CPU-bound programs? (2)

2) Consider a scenario with two processes in a system: Process P1 requires 50 units of CPU time, while Process P2 requires only 10 units. Explain the Convoy effect in the context of this setup. (5)

A)

B) The two processes, P0 and P1, share the following variables:
boolean flag[2]; /* initially false */
int turn;
The structure of process Pi (i == 0 or 1) is defined by the below code snippet, where the other process is Pj (j == 1 or 0).

```
do {
   flag[i] = true;

   while (flag[j]) {
      if (turn == j) {
         flag[i] = false;
         while (turn == j)
            ; /* do nothing */
         flag[i] = true;
      }
   }

      /* critical section */

   turn = j;
   flag[i] = false;

      /* remainder section */
} while (true);
```

(3)

Prove that the above defined 2-Process algorithm satisfies all three requirements for the critical-section problem.

C) Discuss the advantages of using multithreading in programming. Highlight how it enhances performance, efficiency, and resource utilization in various computing environments. (2)

3) Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 2150, and the previous request was at cylinder 1,805. The queue of pending requests, in FIFO order, is: **2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681.**

A) Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? (5)

a) FCFS
b) SSTF
c) SCAN
d) LOOK

B) Consider a system consisting of four resources of the same type that are shared by three processes, each of which needs at most two resources. Show that the system is deadlock free (3)

C) Explain the concept of thrashing in the context of operating system memory management. Describe the conditions that lead to thrashing, its impact on system performance, and the mechanisms that can be employed to mitigate or prevent it. (2)

4) Consider the following snapshot of a system in **Table Q10**:

A)

Table Q10:

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	1	2	0	2	4	3	2	6	1	5	0	2
P1	0	1	1	2	2	4	1	4				
P2	0	2	4	0	3	6	5	1				
P3	1	2	0	1	2	6	2	4				
P4	1	0	0	2	3	1	1	3				

(5)

Answer the following questions using the banker's algorithm:

a) Illustrate that the system is in a safe state by demonstrating an order in which the processes may complete.

b) If a request from process P1 arrives for (1, 1, 0, 1), can the request be granted immediately?

B) With the help of an example and suitable code snippets, explain the solution to the Producer Consumer Problem using Semaphores. (3)

C) With the help of an example explain the concept of Dynamic Relocation in Memory Management. (2)

5) Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)? Rank the algorithms in terms of how efficiently they use memory. (5)

A)

B) With the help of a neat block diagram, explain the different steps involved in handling a Page Fault in an operating system that uses Demand Paging. (3)

C) Briefly explain the drawbacks of the contiguous memory allocation technique. (2)

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